

Registers

- A register is a group of flip-flop which each flip-flop capable of storing one bit information.
- When one flip-flop is connected to another to increase the storage capacity is called Register.
- An n-bit register has a group of n flip-flops and is capable of storing any binary information of n bits.
- The flip-flop hold the binary information and the gates control when & how new information is transferred into the register.

Shift Register:

- A register capable of shifting its binary information in one or both directions is called shift register.
- Example: Left shift and Right shift
- Shift registers are used to implement arithmetic operation.
- All flip-flops receive common clock pulses that initiate the shift from one stage to next stage.
- The storage capacity of a register is the number of bits (1s and 0s) of digital data it can retain, each stage of a shift register represents one bit of storage capacity and therefore the number of storage in a register determines its total storage capacity.
- The shift capability of a register permits the movement of data from stage to stage within the register or into or out of the register upon the application of clock pulse.

PS(Q_n)	NS(Q_{n+1})	D
0	0	0
0	1	1
1	0	0
1	1	1

Types of shift Register:

1. Serial in Serial Out (SISO)
2. Serial in Parallel out (SIPO)
3. Parallel in Serial out (PISO)
4. Parallel in Parallel out (PIPO)

Mode	Clock needed for n-bit Shift Register		
	Loading	Reading	Total
SISO	N	n-1	2n-1
SIPO	N	0	N
PIPO	1	0	1
PISO	1	n-1	n

1) Serial in Serial Out (SISO) Shift Register

SISO shift registers accept data serially i.e., one bit at a time and output taken from it is also in serial form.

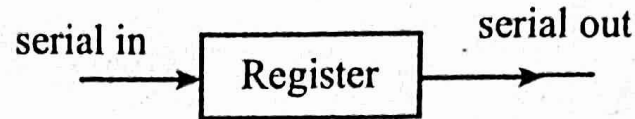


Fig.: Block diagram of right shift register

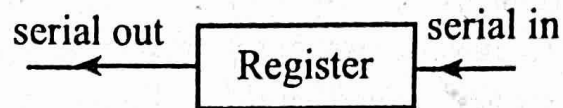


Fig.: Block diagram of left shift register

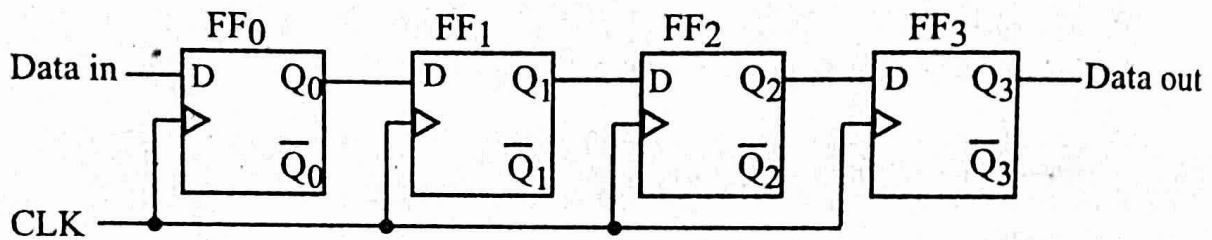


Fig.: Logic diagram of serial in serial out shift register (right shift register).

Let's explain its operation by storing and retrieving "1011". Let the initially the register content is clear i.e., 0000. The rightmost data is entered first.

Clock	Register content				
	Q ₀	Q ₁	Q ₂	Q ₃	
Initially	0	0	0	0	
CLK 1	1	0	0	0	
CLK 2	1	1	0	0	
CLK 3	0	1	1	0	
CLK 4	1	0	1	1	All data stored after 4 th clock
CLK 5	0	1	0	1	
CLK 6	0	0	1	0	
CLK 7	0	0	0	1	
CLK 8	0	0	0	0	Register clear after 8 th clock

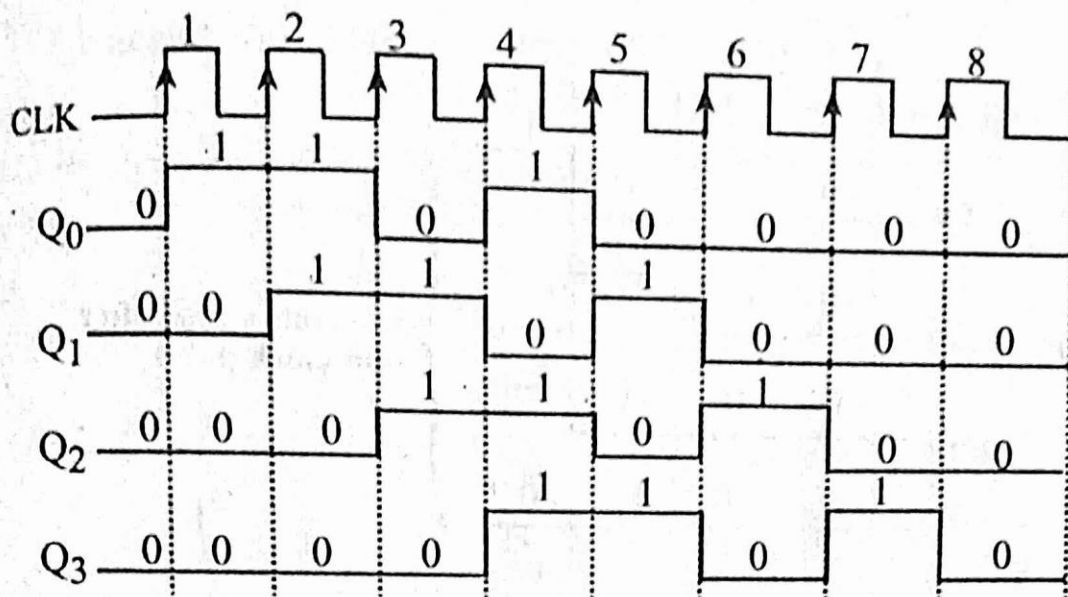


Fig.: Timing diagram for storing 1011

2) Serial in Parallel Out (SIPO) Shift Register

Data are entered serially but output is taken parallelly. Once the data are stored, each bit appears on its respective output line, all at a time.

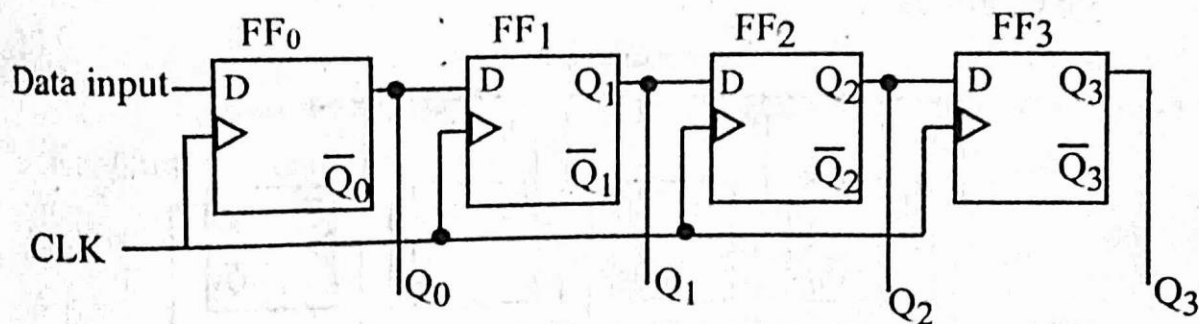


Fig.: Logic diagram of serial in parallel out (SIPO) shift register.

Let 1010 is stored from rightmost bit.

CLK	Register content				
	Q ₀	Q ₁	Q ₂	Q ₃	
Initially	0	0	0	0	Data is completely stored after 4 th clock. Output is available all at a time parallelly.
CLK 1	0	0	0	0	
CLK 2	1	0	0	0	
CLK 3	0	1	0	0	
CLK 4	1	0	1	0	

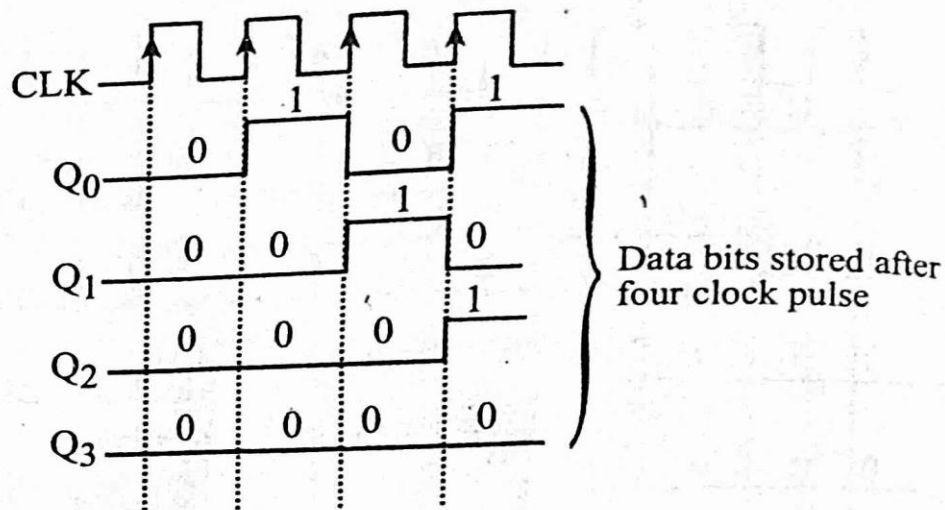


Fig.: Timing diagram for storing 1010 in SIPO shift register.

3) Parallel in Parallel Out (PIPO) Shift Register

Data are entered simultaneously into their respective stages on parallel lines rather than on a bit-by-bit basis. Output is available simultaneously.

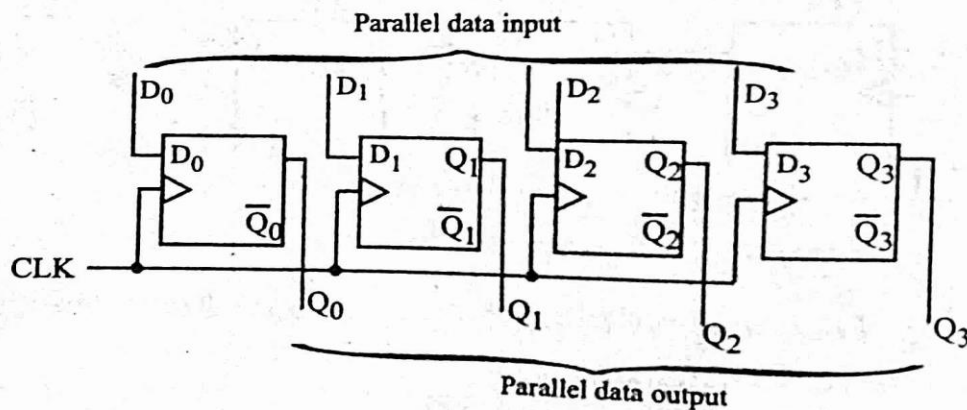


Fig.: Logic diagram of 4-bit parallel in parallel out (PIPO) shift register.

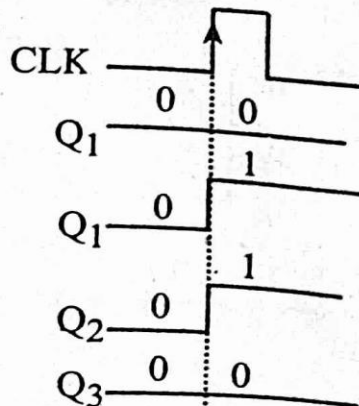


Fig.: Timing diagram for storing 0110 in PIPO shift register

4) Parallel in Serial Out (PISO) Shift Register

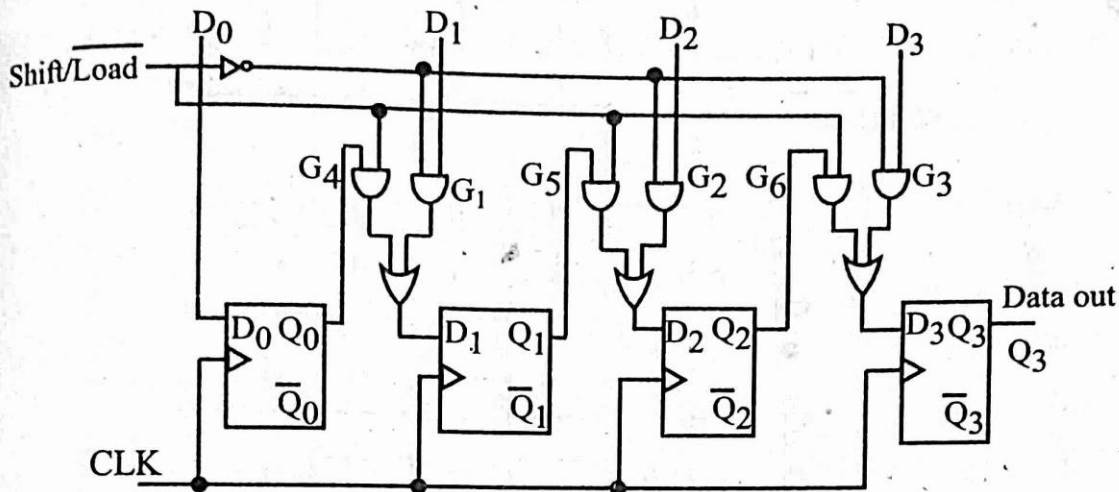


Fig.: Logic diagram of parallel in serial out (PISO) shift register.

Data are entered parallelly but taken in a serial mode. When shift/ Load is LOW; G_1 , G_2 , G_3 gates are enabled so it allows data D_1 , D_2 , D_3 to be applied at D input of flipflops. When shift/ Load is HIGH; G_4 , G_5 , G_6 gates are enabled, allowing bits to shift right from one stage to another on each clock pulse. Output is taken at Q_3 serially.

Let's consider input data as

$D_0 = 1$, $D_1 = 0$, $D_2 = 1$, $D_3 = 0$ (i.e., 1010)

On clock pulse 1, shift/ Load is LOW, so parallel data ($D_0 D_1 D_2 D_3 = 1010$) are loaded into register making Q_3 at 0. On clock pulse 2, shift/ Load is HIGH, so data is shifted i.e., the 1 from Q_2 is right shifted onto Q_3 and so on.

Register content					
CLK	Q_0	Q_1	Q_2	Q_3	
1	1	0	1	0	Data is entered parallelly
2	0	1	0	1	
3	0	0	1	0	
4	0	0	0	1	
5	0	0	0	0	

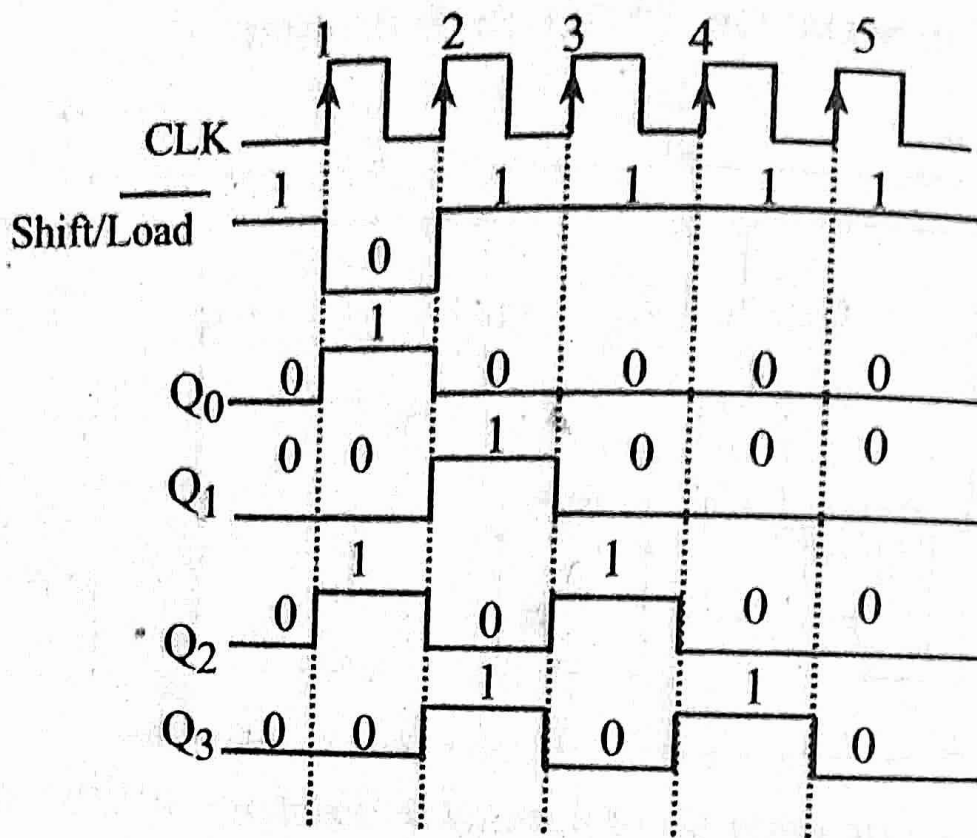


Fig.: Timing diagram for storing 1010 in PISO shift register

Four bit data 1010 is entered into serial in parallel out shift register. Draw the circuit diagram and output waveforms after 1, 2, 3, 4 clock pulses. [2064 Jestha]

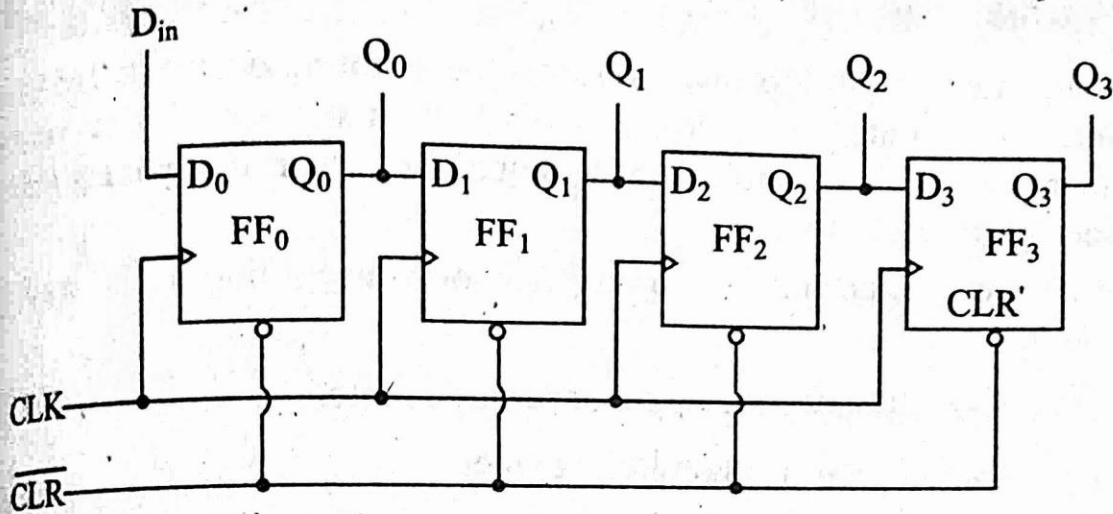
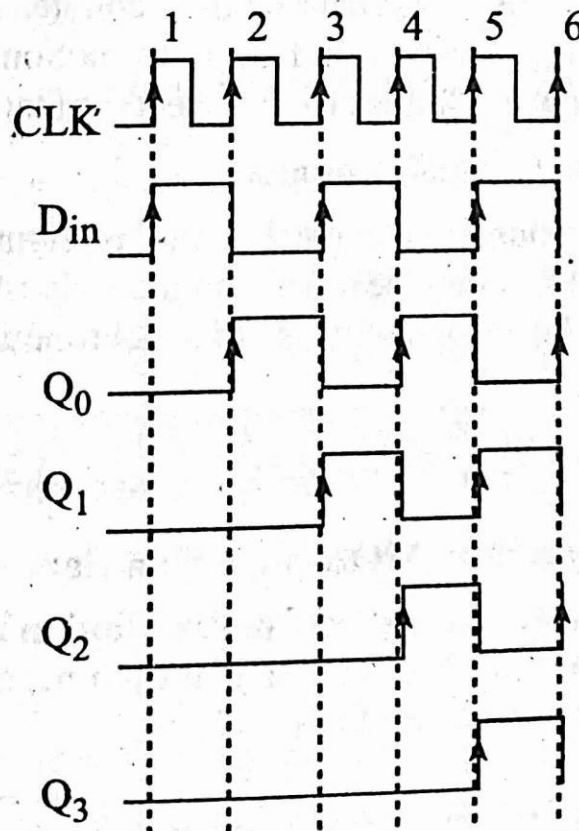
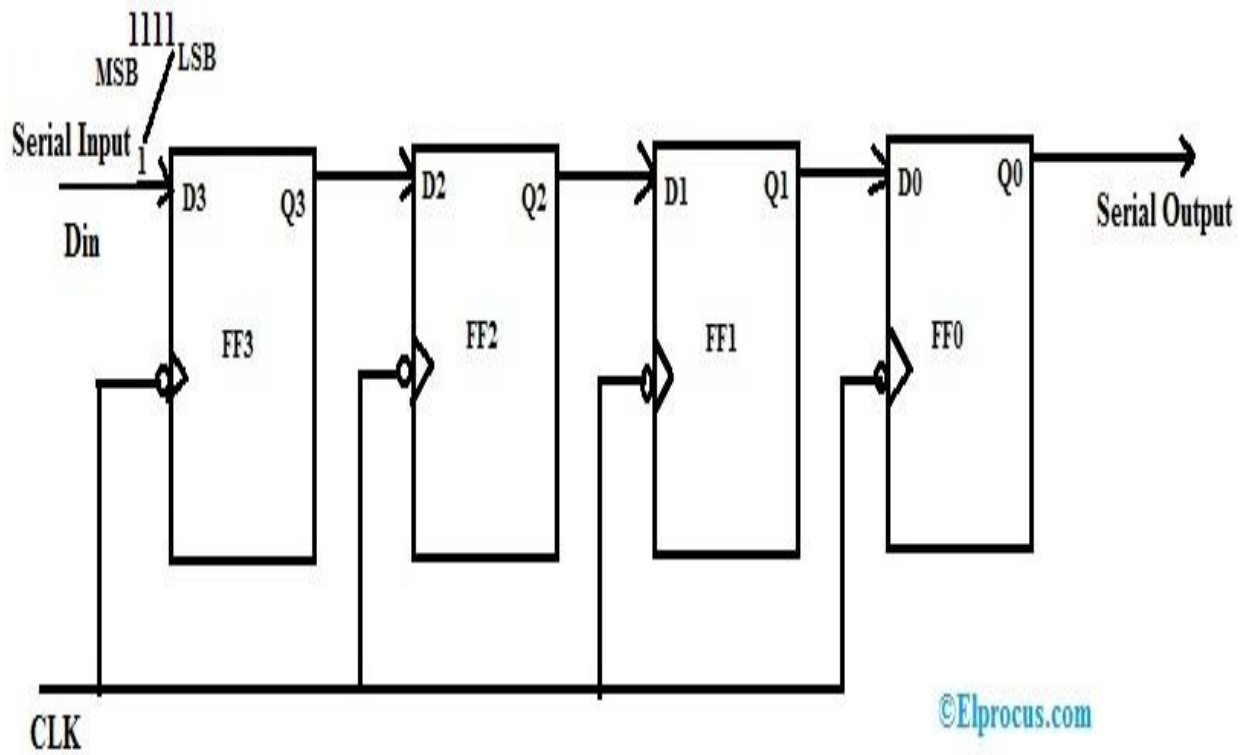
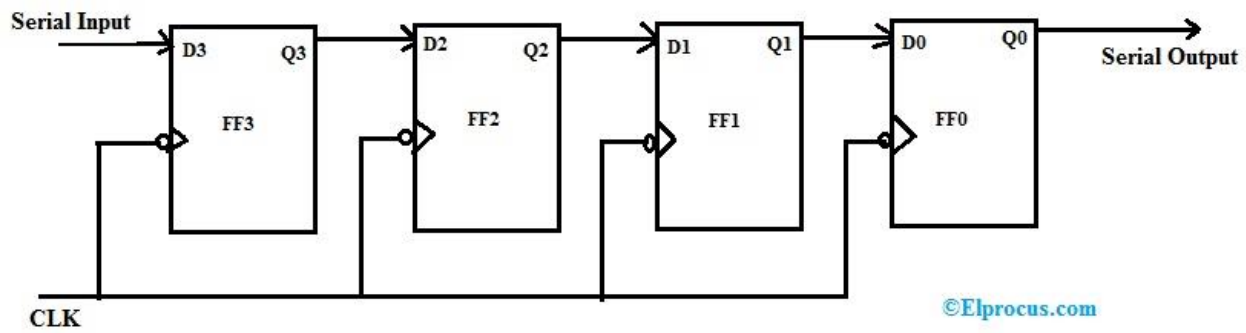


Fig.: 4-bit serial in parallel out shift register

Now, timing diagram for above circuit is



Serial in Serial out (SISO):(with clock pulses=0)



The **SISO** shift register truth table is shown below.

CLK	'Q3'	'Q2'	'Q1'	'Q0'
Initially (Reset)	0	0	0	0
1 st Falling Edge	1	0	0	0
2 nd Falling Edge	1	1	0	0
3 rd Falling Edge	1	1	1	0
4 th Falling Edge	1	1	1	1

